

database, the result of the SQL command will always be true (which means the system recognized the illegal entry into the system as a genuine user). This will open the doors for the hacker to gain access

A login form with a 'Name' field containing the character 't', an empty 'Password' field, and a 'Login' button.

to the web server. The hacker can run similar attacks on the database to either delete the data or insert dummy data into the database.

Most of the website programming languages allow developers to handle user inputs used in an SQL query. SQL injection attacks can be prevented by filtering user input.

Automated tools to detect SQL Injections

Different automated tools can help you perform SQL injection attacks and detect whether your web application is safe or not. SQLMAP is one of the best available tools to detect SQL injection attacks. Once you have downloaded the tool, get the header request from the infected page and run the `./sqlmap.py` command in the SQL. The SQLMAP tool will test and show the result whether there is any vulnerability or not.

Few of the other automated tools to detect SQL Injection include SQLS-mack, SQLPing2, SQLNinja, JSQ Injection.

XSS vulnerabilities

Similar to SQL Injection, most hackers take advantage of the Cross-Site Scripting (XSS) to hack websites and web applications. With the XSS, attackers get to inject malicious client-side JavaScript into web pages that are viewed by users. With XSS, attackers can hijack user sessions, force a website redirect to another page/website, and so on.

For instance, in a web application that is vulnerable to attacks, when a specific HTML code is generated dynamically, and the user input is directly reflected on the page, it becomes easy for the attacker to insert his custom code into the website server. Doing this will grant the attacker access to all the other pages on the same domain and the ability to read the data in the page or even set customized cookies that can store the session information. Using these cookies, the attacker can log into the